CGEC

Center for Genomic Experimentation & Computation
Molecular Sciences Institute

First Annual Alpha Project Retreat

Costanoa Coastal Lodge & Camp

2001 Rossi Road at Hwy 1
Pescadero, CA 94060
650-879-1100
http://www.costanoa.com

Location and Directions

Costanoa is located about an hour (55 miles) south of San Francisco, off Highway 1.

Heading South on Highway 1

If you are traveling south on Highway 1, the last major town you will pass through is Half Moon Bay. At this point you are about 26 miles from Costanoa. 10.5 miles before you reach Costanoa you will see signs for the historic town of Pescadero. Do not turn, continue south on Highway 1. The next major landmark you will see is the majestic Pigeon Point Lighthouse. This distinctive structure is located on the ocean side of the highway and is clearly visible. As you drive past the lighthouse you are just under 3 miles from reaching Costanoa. After passing a large grove of Eucalyptus trees on your left, watch for the Costanoa signs and the only left hand turn lane from the highway. Turn left onto our entrance road (Rossi Road) and continue until you reach the entrance gate. From the entrance gate follow directions to your appropriate check-in area.

Heading North on Highway 1

If you are traveling north on Highway 1, the last major town you will pass through is Santa Cruz. At this point you are about 20 miles from Costanoa. 12 miles before Costanoa you will pass through the town of Davenport. Continue north on Highway 1. About 2 miles before you reach Costanoa you will see signs and an entrance for Año Nuevo State Reserve on your left. Pass the reserve and begin watching for Costanoa entrance signs on your right about two miles down the road. Turn right onto our entrance road (Rossi Road) and continue until you reach the entrance gate. From the entrance gate follow directions to your appropriate check-in area.

From the San Jose area

Travel southwest on Highway 17 towards Santa Cruz and the Pacific Ocean. Near Santa Cruz follow signs for Highway 1 North. Begin traveling north on Highway 1. At this point you are about 20 miles from Costanoa. 12 miles before Costanoa you will pass through the town of Davenport. Continue north on Highway 1. About 2 miles before you reach Costanoa you will see signs and an entrance for Año Nuevo State Reserve on your left. Pass the reserve and begin watching for Costanoa entrance signs on your right about two miles down the road. Turn right onto our entrance road (Rossi Road) and continue until you reach the entrance gate. From the entrance gate follow directions to your appropriate check-in area.

Activities

THE SPA @ COSTANOA
Discover the relaxing pace of nature with one of Costanoa's Full, Deluxe or Luxurious Sessions, available as a Swedish, Deep Tissue or Shiatsu massage. Add Aromatherapy or Arnica, to any of those treatments. Let our skilled therapists indulge you with a well-deserved massage after that long hike or exhilarating bike ride. Or, simply enjoy a little pampering as part of your coastal escape. All massages are by appointment only. MAKE YOUR APPOINTMENTS IN ADVANCE BY CALLING: 650.879.1100, x7413 or go online.
http://www.costanoa.com/spa.html

HIKING
For a trail map, go to:
http://www.costanoa.com/trailmap.html

YOGA
Stretch your mind and body with one of our weekend YOGA classes. YOGA takes place every Sunday morning from 10-11am, Memorial Day through Labor Day, in the Ranch House Community Room. Each class is $15 per person. Private sessions may be arranged through the Adventure Experience Manger, Oscar Allan at **650.879.1000** or by email at oallan@jdvhospitality.com

MOUNTAIN BIKE RIDING
Hit the trail with one of our TREK 6500 front suspension Mountain Bikes. Available for rent everyday at the Activities Desk at the Ranch House for $15/hour or $45 for a full day.

HORSEBACK RIDING
Explore our 400 acres of coastal grasslands with our resident Horseback riding service Blue Sky Riding Experience. Reservations available through the Front Desk by calling 650.879.1100 x7428. Horseback riding is available year round weather conditions permitting. Trips for groups of 4 or more can be arranged by appointment.

INTERPRETIVE HIKES
Discover the local flora and fauna with one of our resident naturalists; interpretive walks offered every weekend year round. Hikes are free of charge and meet at the Ranch House Fridays 2-4pm and

Saturdays and Sundays 10-12 noon. Check the activities board in the Ranch House lobby to confirm that the times are accurate for the day you are visiting.

NEARBY STATE PARKS
Explore any of the nearby State parks on foot or bike:

- **BIG BASIN STATE PARK**

Over 1900 acres of parkland including redwoods, waterfalls, and hiking trails

- **BUTANO STATE PARK**

A hidden gem in the Santa Cruz Mountains just up the road from Costanoa

- **ANO NUEVO STATE RESERVE**

Elephant seals and bird life abound

- **PIGEON POINT LIGHTHOUSE**

The weekend tours take you to the top of the second highest lighthouse on the west coast.

- **NATURAL BRIDGES**

This beach, with its famous natural bridge, is an excellent vantage point for viewing shore birds, migrating whales, and seals and otters playing offshore.

OTHER DAY TRIPS FROM COSTANOA

BONNY DOON VINEYARD
Enjoy the award winning wineries of the Santa Cruz Mountains www.bonnydoonvineyard.com

KAYAKING
Take advantage of the coastal kayaking opportunities along the central coast.
http://www.kayaksantacruz.com/

INDOOR ROCK CLIMBING GYM
http://www.pacificedgeclimbinggym.com/

WHALE WATCHING AND FISHING
http://www.stagnaros.com/

SAILING
http://www.pacificsail.com/

Agenda

Friday, November 21, 2003

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12 noon – 6:00 pm	**Check-In and Free Time**
6:00 pm - 6:30 pm	*Meet & Greet and Cocktails* Redwood / Oak Room (located in main lodge)
6:30 pm - 7:30 pm	**Dinner** Redwood / Oak room (located in main lodge)
7:30 pm - 8:30 pm	DINNER TALK *Title TBA* Roger Brent President and Research Director, Molecular Sciences Institute Director, Center for Genomic Experimentation & Computation Adjunct Professor, Department of Biopharmaceutical Sciences, University of California, San Francisco 2003 Gabby Award Recipient

Saturday, November 22, 2003

8:30 am - 9:00 am	**Breakfast** Redwood / Oak Room (located in main lodge)

INTRODUCTION: 9:00 am – 10:45 am
(Redwood / Oak Room located in main lodge)

9:00 am - 9:15 am	*Welcome and Introductions* Orna Resnekov Molecular Sciences Institute
9:15 am - 10:00 am	*History, Motivation, and Overview of the Alpha Project* Roger Brent Molecular Sciences Institute
10:00 am - 10:45 am	*The Problem of Prediction as it Pertains to Biology* Drew Endy Department of Biology and Division of Biological Engineering Massachusetts Institute of Technology
10:45 am - 11:00 am	**Coffee Break**

MOLECULIZER, MODELS, AND SOFTWARE SESSION: 11:00 am – 12:00 noon

(Redwood / Oak Room located in main lodge)

11:00 am - 11:30 am	*Moleculizer* Larry Lok Molecular Sciences Institute
11:30 am - 11:40 am	*Model Modules* Ty Thompson Division of Biological Engineering Massachusetts Institute of Technology
11:40 am - 11:50 am	*Monod* David Soergel University of California, Berkeley and Molecular Sciences Institute
11:50 am - 12 noon	*Monod - Demonstration* Jay Doane and Kirindi Choi Molecular Sciences Institute
12 noon - 1:00 pm	**Lunch** Redwood / Oak Room (located in main lodge)

EXPERIMENTAL DATA SESSION: 1:00 pm – 2:10 pm

(Redwood / Oak Room located in main lodge)

1:00 pm - 1:30 pm	*Physiology and Cell Biology of Pheromone Response* Alejandro Colman-Lerner Molecular Sciences Institute
1:30 pm - 1:40 pm	*FRET Analysis* Richard Yu Molecular Sciences Institute
1:40 pm -1:50 pm	*Phosphopeptide Analysis* Robert Maxwell Pacific Northwest National Laboratories
1:50 pm - 2:00 pm	*Quantification of Pathway Components* Kirsten Benjamin Molecular Sciences Institute
2:00 pm - 2:10 pm	*Does Size Matter? Cell Size Control by the Pheromone Mating Pathway* Eduard Serra Molecular Sciences Institute
2:10 pm - 2:30 pm	**Coffee Break**

BREAK-OUT SESSION I – Predictive Models

(Redwood / Oak Room located in main lodge)

2:30 pm - 3:00 pm	Break out into groups and answer the following questions: Why do we need a predictive model? At what level of resolution do we need the model? What is the predictive model good for?
3:00 pm - 4:00 pm	*Break-out Report* Moderator: Shuki Bruck Computation and Neural Systems and Electrical Engineering California Institute of Technology
4:00 pm - 5:30 pm	**Break and Free Time**
5:30 pm - 6:30 pm	**Dinner** Redwood / Oak Room (located in main lodge)

BREAK-OUT SESSION II – Quantitative Data

(Redwood / Oak Room located in main lodge)

6:30 pm - 7:15 pm	Break out into groups and answer the following question: Suppose we had all the quantitative data that we say we need, what do we do with it?
7:15 pm - 8:30 pm	*Break-out Report* Moderator: Drew Endy Department of Biology and Division of Biological Engineering Massachusetts Institute of Technology
8:30 pm	**Adjourn for the evening**

Sunday, November 23, 2003

8:30 am - 9:00 am	**Breakfast** Redwood / Oak Room (located in main lodge)

EXPERIMENTAL PROGRESS SESSION: 9:00 am – 10:40 am

(Redwood / Oak Room located in main lodge)

9:00 am - 9:10 am	*Revisting the Physics Models Underlying Discrete Reaction Event Simulators* Sylvain Olier Department of Biology Massachusetts Institute of Technology

9:10 am - 9:20 am	*Nuclear Export-based Report of Kinase Substrate Motifs* Samantha Sutton Division of Biological Engineering Massachusetts Institute of Technology
9:20 am - 9:30 am	*Light-mediated Activation/Inactivation of Protein Localization and Protein-Protein Interactions* Francois St. Pierre Department of Biology Massachusetts Institute of Technology
9:30 am - 9:40 am	*Single Cell Measurements Meets Genetics: High-Throughput Screens for Signal Transduction Mutants* Gustavo Pesce Molecular Sciences Institute
9:40 am - 10:00 am	**Coffee Break**
10:00 am - 10:10 am	*New Approach for Identifying and Quantifying Modifications of Pathway Components* Orna Resnekov Molecular Sciences Institute
10:10 am - 10:20 am	*What Proteins Can Do* Ximena Ares Molecular Sciences Institute
10:20 am - 10:30 am	*Comparative Analysis of Yeast's Alpha and Secretory Pathways* Michael Gonzales Molecular Sciences Institute
10:30 am - 10:40 am	*Pathway Analysis Across Yeast Species* Ilya Sytchev Department of Biology Massachusetts Institute of Technology
10:40 am - 10:50 am	*Tests of Anti-modularity and Persistent Design* Jeff Gritton Department of Biology Massachusetts Institute of Technology
10:50 am - 11:00 am	**Coffee Break**

EMERGING TECHNOLOGIES SESSION: 11:00 am – 12:30 pm

(Redwood / Oak Room located in main lodge)

11:00 am - 11:30 am	*Sophisticated Instrument Systems to Enable Complex Protocols* Mark Holl Department of Engineering University of Washington, Seattle

11:30 am - 12 noon	*Ultra-Sensitive Analyte Quantification* Ian Burbulis Molecular Sciences Institute
12 noon - 12:30 pm	*Yeast Affinity Reagent Platform* Michael Feldhaus Pacific Northwest National Laboratories
12:30 pm - 1:30 pm	**Lunch** Redwood / Oak Room (located in main lodge)

BREAK-OUT SESSION III – New Technologies
(Redwood / Oak Room located in main lodge)

1:30 pm - 2:00 pm	Break out into groups and answer the following questions: What kind of technologies/abilities would we like to have for the Alpha Project? How do we get them? What experiments or analyses would we like to be able to do? What do we need to be able to do them?
2:00 pm - 3:00 pm	*Break-out Report* Moderator: Roger Brent Molecular Sciences Institute
3:00 pm - 3:10 pm	*Wrap-Up* Roger Brent Molecular Sciences Institute
3:10 pm	**Adjourn**